Posters and Presentations

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Session Outline

- 20 minutes – overview of report options
- Remaining time – group or individual work on reports.

Presentation Tips

- Introduce yourself
- Try to introduce topic for general audience
- Consider memorizing the first 1-2 minutes of your talk
- Aim for about 1 minute per slide and leave time for questions
- Use math formulas as “figures” – do not try to present detailed computations or proofs
- Be less precise than your advisor wants you to be

Poster tips

- Try to go to poster session even if you aren't presenting
- Consider preparing a short (<60 seconds) and medium (2-3min) explanation of your project
- Most people will skim the text on the poster and look more carefully at figures
- Different attendees will interact with you in different ways... poster sessions can be awkward

Microsoft Office

Posters and presentations can be prepared using Microsoft PowerPoint. UWM print and copy services has some examples of posters that you can refer to.

Pros

- Intuitive, graphical interface

Cons

- Writing math is painful
- Office suite costs $$
- Big projects get laggy

LaTeX

Pros

- Highly flexible math support
- Mature ecosystem for bibliography management, references…
- Excellent typesetting engine

Cons

- Accessibility issues
- Steep learning curve
- Overleaf free tier may not compile large projects
- Large projects compile slowly

Realistically, math academics will need to learn LaTeX.

LaTeX Presentations

For presentations, typically the “beamer” document class / package is used.

Example: slides/beamer/liam-proposal

Overleaf Beamer Documentation

Beamer User Guide

Beamer slide tips

- Decide on theme early
- Consider ad hoc usage of e.g. `\footnotesize` or `\scriptsize` in slides if you must reduce math size
- I recommend `subcaption` package for nested figures
- Use transparent background for images if possible

LaTeX Posters

LaTeX posters can be created with "beamerposter" or "tikzposter"; I've mostly used the former.

Examples: posters/beamer/npbic, posters/beamer/kelp-detection

beamerposter on CTAN

Overleaf Poster documentation

Beamerposter tips

- Decide on theme early
- Leave `\documentclass[10pt]{beamer}` but change `\usepackage[orientation=landscape,height = 40, width = 60, scale=1.25]{beamerposter}`
- See npbic poster for custom poster header example
- Set `\setbeamertemplate{caption}[numbered]` in preamble to force numbering of figures
- Set `\setbeamertemplate{navigation symbols}{}` in preamble to hide navigation symbols

Jupyter Notebook Conversion

A nice approach for projects involving code, especially if you want to show code blocks, is to write a jupyter notebook and then use pandoc/nbconvert/quarto/etc to convert to a pdf or web format.

Pros

- Good accessibility support
- Conversion to a variety of final formats
- Better reproducibility of code output

Cons

- Typesetting is ok
- Limited math support
- Some formatting issues in pdf output

Quarto

A scientific report publishing tool supporting notebook conversion, with good defaults with a bunch of target formats.

Examples: slides/quarto/finite-elements (slides), posters/quarto/nps (poster)

Typst

Is an aspiring LaTeX alternative written in rust. This presentation is written in typst.

Pros

- Good accessibility support
- Conversion to a variety of final formats
- Very fast

Cons

- Typesetting is good (and improving)
- Some math support

Questions?

Work time